In general, when a linear operator acts on a vector, it will change the direction of the vector. This means that what comes out of the machine will not just be the input vector multiplied by a number. But for a particular linear operator, there will be certain vectors whose directions are the same when they come out as they were when they went in. These special vectors are called *eigenvectors*. The definition of an eigenvector of **M** is a vector |λ> such that

$$\mathbf{M}|\lambda\rangle = \lambda|\lambda\rangle. \qquad (3.5)$$

The double use of λ is admittedly a little confusing. First of all, λ (as opposed to |λ>) is a number—generally a complex one, but still a number. On the other hand, |λ> is a ket-vector. Furthermore, it is a ket with a very special relationship to **M**. When |λ> is fed into the machine **M**, all that happens is that it gets multiplied by the number λ. I'll give you an example. If **M** is the 2 × 2 matrix

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix},$$

then it's easy to see that the vector

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

just gets multiplied by 3 when **M** acts on it. Try it out. **M** also happens to have another eigenvector:

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$